

Method	Size	Grading	Both
Base ZS	36.1%	34.3%	12.0%
GPT-4.1 FS	62.0%	59.3%	29.6%
Direct LoRA	89.8%	78.7%	71.3%
CoT-Aug	91.7%	86.1%	79.6%

Table 5: Granulometry results by axis.

Class (N)	Direct	CoT-Aug
pitted_surface (60)	83%	85%
scratches (60)	63%	85%
crazing (60)	83%	68%
patches (60)	52%	75%
rolled-in_scale (60)	58%	55%
inclusion (60)	38%	32%
Overall	63.1%	66.7%

Table 7: Per-class accuracy, steel surface.

Class (N)	Direct	CoT-Aug
no_defect (60)	87%	98%
porosity (60)	83%	83%
lack_of_pen. (60)	65%	72%
cracks (60)	58%	50%
Overall	73.3%	75.8%

Table 6: Per-class accuracy, weld defects.

Class (N)	Direct	CoT-Aug
spheroidite (74)	74%	72%
network (20)	75%	80%
spher.+widm. (15)	13%	27%
pearl.+spher. (5)	100%	80%
pearlite (3)	67%	100%
Overall	65.8%	66.7%

Table 8: Per-class accuracy, UHCS.

The challenge here is compound classes. “Spheroidite+widmanstatten” requires detecting two co-existing features in the same image. With only 6 training images for this class, both approaches struggle. The augmented method doubles the accuracy on this class (13% to 27%), showing that contrastive descriptions help the model learn compound visual patterns. The small test set sizes for pearlite (3) and pearlite+spheroidite (5) make per-class comparisons on those classes unreliable.

6 Discussion

6.1 When Does CoT Distillation Help Most?

The improvement from answer-conditioned CoT distillation varies across tasks: +8.3pp on granulometry, +3.6pp on steel, +2.5pp on weld defects, and +0.9pp on UHCS. This variation correlates with task structure.

The largest gain (granulometry) occurs on a task with many classes (9) and fine-grained visual distinctions. Grading requires understanding spatial relationships between particles (whether gaps are empty, partially filled, or completely filled). This kind of relational reasoning transfers well through natural language descriptions. Direct JSON training cannot teach these relationships explicitly.

The smallest gain (UHCS) occurs on a task dominated by a majority class (spheroidite: 62% of test set) and compound classes that require detecting multiple co-existing features. When the test set is imbalanced and the hard classes have very few test samples, the method’s advantage on minority classes is diluted in the overall accuracy.

6.2 Why Answer-Conditioning Is Necessary

Answer-conditioning is not optional. The frontier model’s own classification accuracy ranges from 29.6% (granulometry) to 91.1% (steel surface, with few-shot references). If we used standard CoT distillation (let the teacher classify and explain), between 9% and 70% of the